

United States GDP Regimes, Fiscal Structure, and Distributional Growth

A Reproducible Data-Science Report, 1920–Present

Diogo Ribeiro
`us-gdp-regime-1920`

July 10, 2026

Abstract

This report summarizes a reproducible empirical workflow for long-run United States real GDP growth from 1920 onward. The analysis combines a Maddison-based real GDP proxy, FRED/BEA validation data, piecewise growth-regime detection with formal structural-break inference, federal fiscal context, tax-regime event dynamics, and distributional measures linking GDP per capita to real worker earnings. The headline statistical result is that a single long-run trend fits the log GDP level well in-sample, but the level is statistically indistinguishable from a unit-root process, so annual growth — which is stationary — is the appropriate object for regime analysis. Growth is best read as a sequence of regimes, and a sequential bootstrap break test confirms the number of regimes that the information criterion selects. The distributional result is equally important: aggregate real GDP per capita rose much faster than real median weekly earnings after the late 1970s, so GDP growth alone is an incomplete proxy for broad buying power. Fiscal and tax variables add useful context, but the report distinguishes descriptive timing, lagged association, and causal identification throughout.

Keywords: United States GDP, growth regimes, structural breaks, public debt, tax policy, real wages, distributional analysis, local projections, reproducible research

All generated tables and figures are reproducible from the Python pipeline in this repository. Every statistic in the text is a machine-generated macro; raw downloaded data are intentionally excluded from version control.

1 Executive Summary

The project began with a simple question: whether the United States grew at one roughly constant speed after 1920, or whether the data support distinct growth regimes. The answer is mixed in the way macroeconomic time series usually are: the log level of real GDP has a strong long-run trend, but the annual growth process is segmented by depression, war, demobilisation, and a long postwar period.

The fitted log-trend model implies an annualized trend growth rate of approximately 3.16% with an in-sample R^2 of 0.987. That R^2 should not be read as evidence for a deterministic trend:

augmented Dickey-Fuller and KPSS tests both indicate that log real GDP is consistent with a unit root, so a straight-line trend mechanically fits a persistent series well. Annual growth, by contrast, is stationary. The piecewise growth model selects 5 broad regimes in the Maddison-derived annual GDP series (Table 1), and a sequential parametric-bootstrap break test agrees that additional regimes beyond this point are not statistically supported. These regimes are descriptive statistical summaries; they are not automatic causal explanations.

The expanded fiscal and distributional layers change the interpretation. Federal debt, deficits, and tax laws help structure the historical narrative, but they are endogenous to the same shocks that affect GDP. More importantly, GDP per capita and real worker earnings diverged strongly after the index base year 1979. Indexed to 1979, real GDP per capita was about 218.4 by 2026 (more than double its base-year level), while real median weekly earnings were about 113.4. In other words, aggregate output per person more than doubled, while the median weekly earnings index rose only modestly above its base.

2 Data and Reproducibility

2.1 Core GDP Series

The primary GDP series is built from the Maddison Project Database 2023 by multiplying United States real GDP per capita by population. This produces a long-run real GDP proxy suitable for growth-rate and log-trend analysis from 1920 onward. The modern validation source is FRED/BEA annual real GDP (GDPCA), available from 1929 onward.

The source-validation table compares Maddison-derived growth with FRED/BEA growth from 1930 to 2022. The overlap has 93 annual observations, a growth-rate correlation of 0.955, a mean absolute difference of 0.63 percentage points, and an RMSE of 1.47 percentage points. This is close enough for regime-level analysis, while still reminding the reader that Maddison historical estimates and BEA national accounts are not the same object.

2.2 Fiscal, Tax, and Distributional Data

The fiscal context layer uses FRED/OMB federal debt, receipts, outlays, deficit, and interest-outlay ratios as percent of GDP. The tax-regime event layer uses a curated catalog of broad federal tax-law changes, with signed ordinal shock values and narrative classifications. The distributional layer adds BEA real GDP per capita, BLS real median weekly earnings, BLS real hourly compensation, BEA federal receipt composition, IRS top and bottom statutory income-tax rates, and BLS Consumer Expenditure federal income-tax rates by income quintile.

These series answer different questions. GDP per capita measures aggregate real output per person. Median weekly earnings measure the real earnings of the median full-time wage and salary worker. Federal income-tax rates by Consumer Expenditure income quintile measure average federal income-tax burdens for consumer units, not the full incidence of the tax-and-transfer system.

3 Methods

3.1 Trend Model and Unit-Root Diagnostics

The long-run trend is estimated on log real GDP:

$$\log(GDP_t) = \alpha + \beta \cdot year_t + \epsilon_t. \quad (1)$$

The annualized trend growth rate is computed as $\exp(\beta) - 1$. Because annual macroeconomic series are highly autocorrelated, the slope is reported with Newey-West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors [4], using 4 lags on 103 observations. To guard against a spurious-trend reading, the log level and the growth series are each tested with the augmented Dickey-Fuller test [5], whose null is a unit root, and the KPSS test [6], whose null is stationarity. Reporting both avoids over-reading a single test.

3.2 Piecewise Growth Regimes and Break Inference

Growth regimes are estimated on annual percent changes:

$$growth_t = \mu_j + u_t, \quad t \in \text{regime } j. \quad (2)$$

The algorithm searches over candidate breakpoints with dynamic programming and selects the segmentation that minimizes a Gaussian information criterion (BIC by default), counting each segment mean, each break location, and the common residual variance as free parameters. A segment is labelled “above mean” if its mean growth rate exceeds the full-sample mean. This is a descriptive label, not a welfare or causality statement.

The number of regimes is not taken on faith from the criterion. Each additional break is tested with a sequential $\sup F(l+1|l)$ statistic in the spirit of Bai and Perron [1, 2]. Because the break date is estimated, the statistic does not follow a standard F distribution, so the p-value is obtained by a parametric bootstrap (199 replications) that simulates data from the fitted smaller model. Uncertainty in the break *locations* is quantified separately by a residual bootstrap (499 replications) that re-segments each resample and reports percentile intervals for every break year.

3.3 Dynamic Tax Effects

The dynamic tax-effects layer uses two complementary designs. First, local projections [3] estimate the association between a signed tax shock and future GDP growth at each horizon:

$$growth_{t+h} = \alpha_h + \beta_h shock_t + \gamma_h growth_{t-1} + \gamma'_h growth_{t-2} + e_{t+h}. \quad (3)$$

Because the horizon- h local-projection residual is serially correlated up to order h by construction, the Newey-West lag length grows with the horizon rather than being fixed. Second, distributed-lag regressions include current and lagged tax shocks in a single annual growth equation. Both designs estimate delayed dynamics, but they do not solve identification by themselves. The report therefore separates all catalogued tax events from a smaller subset classified as plausibly exogenous long-run reforms.

3.4 Distributional Analysis

The distributional layer indexes real GDP per capita, real median weekly earnings, and real hourly compensation to a common base year:

$$index_{x,t} = 100 \times \frac{x_t}{x_{base}}. \quad (4)$$

The main gap measure is:

$$gap_t = index_{GDPpc,t} - index_{medianearnings,t}. \quad (5)$$

Tax-burden shift is approximated with three sets of proxies: receipt composition, statutory top-bottom income-tax rate spread, and Consumer Expenditure federal income-tax rates by income quintile. These proxies do not constitute a full incidence model, but they directly address whether the tax structure moved away from more progressive bases and whether this aligns with GDP growth or worker earnings.

4 Results

4.1 Long-Run GDP Trend and Growth Regimes

Figure 1 shows the log real GDP path and fitted linear trend. The in-sample fit is high ($R^2 = 0.987$) and the slope is precisely estimated even under HAC standard errors (slope $t = 36.3$). That precision, however, is exactly what a near-integrated series produces: the augmented Dickey-Fuller test does not reject a unit root in log real GDP (ADF statistic -0.11 , $p = 0.993$), and the KPSS test rejects trend-stationarity ($p = 0.037$). The annual growth series behaves the opposite way — ADF rejects a unit root ($p < 0.001$) and KPSS does not reject stationarity ($p = 0.100$). This is the formal justification for analysing regimes on growth rather than on the level.

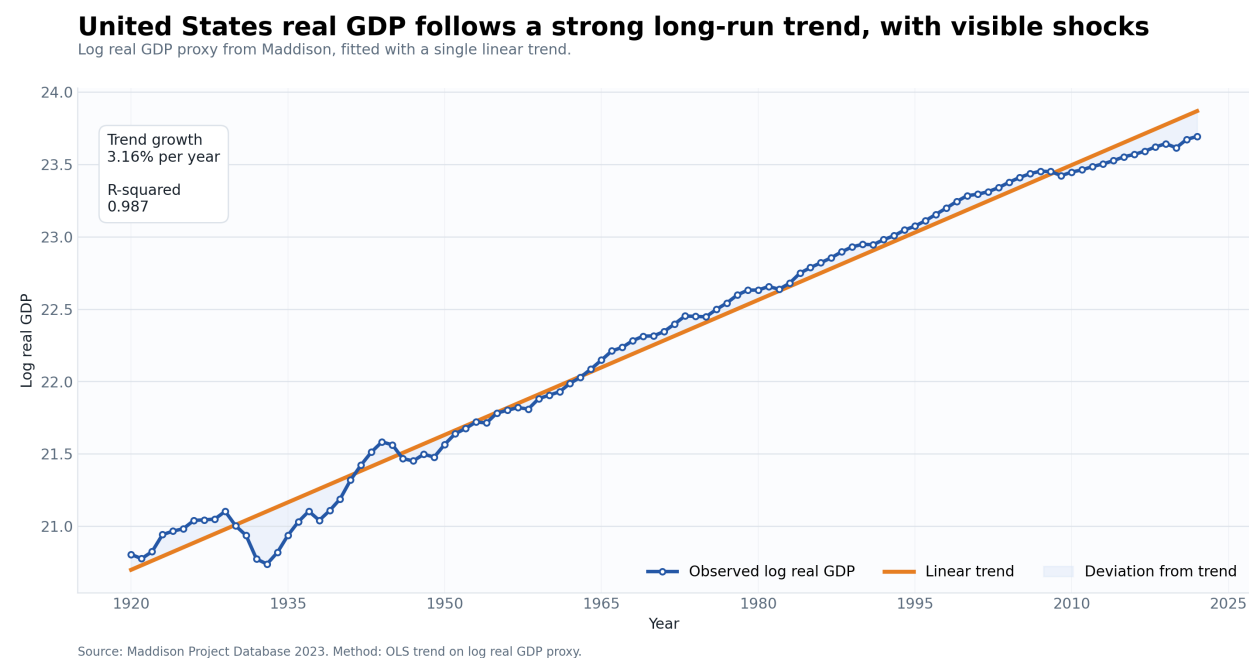


Figure 1. Log real GDP trend from the Maddison-derived United States real GDP proxy. The fitted trend summarizes average exponential growth, while deviations identify periods that require historical interpretation.

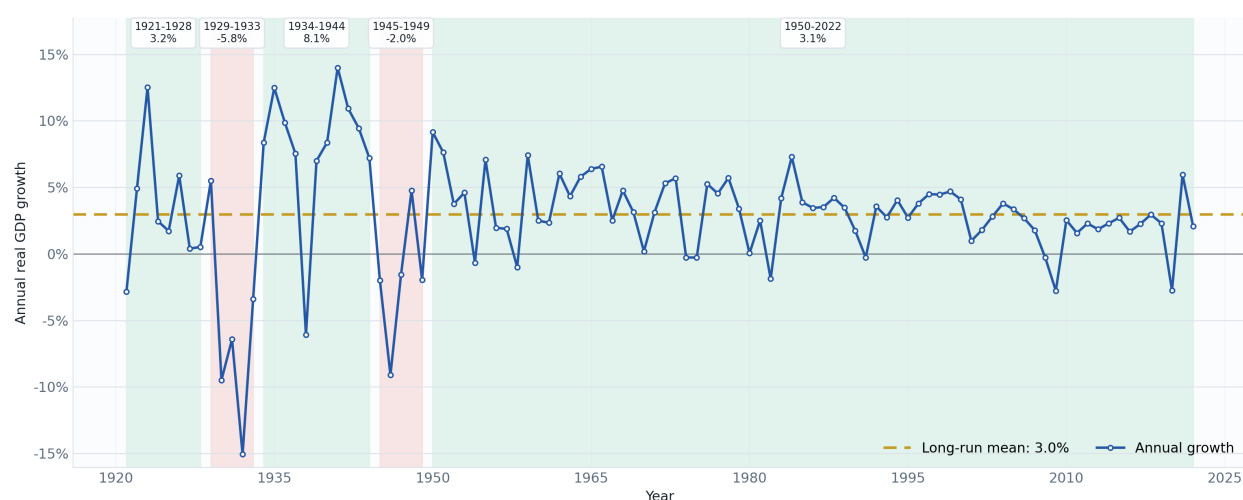
Table 1 summarizes the fitted growth regimes. The severe 1929–1933 contraction and the 1934–1944 rebound/mobilisation period dominate the early sample. The post-1950 period is long and relatively stable by this model’s penalty criterion, with mean annual growth of approximately 3.11% against a full-sample mean of 2.97%.

Table 1. Fitted annual GDP growth regimes

Start	End	Years	Mean growth	Label
1921	1928	8	3.19%	Above mean
1929	1933	5	-5.78%	Below mean
1934	1944	11	8.10%	Above mean
1945	1949	5	-1.98%	Below mean
1950	2022	73	3.11%	Above mean

Growth regimes separate temporary volatility from sustained phases

Piecewise mean growth segments selected on annual real GDP growth.



Source: Maddison Project Database 2023. Green/red bands indicate above/below full-sample mean segment growth.

Figure 2. Piecewise annual real GDP growth regimes. Bands are statistical segments, not claims that one historical event caused the entire segment.

4.2 Are the Breaks Statistically Significant?

Table 2 reports the sequential break tests. Reading down the table, each additional regime up to the selected model earns a small bootstrap p-value, while the first insignificant step marks the point where the data stop supporting further segmentation. This is the formal counterpart of the information-criterion choice: the criterion and the test agree on the number of regimes.

Table 2. Sequential $\sup F(l+1|l)$ break tests with parametric-bootstrap p-values

Segments (null)	Segments (alt)	F statistic	Bootstrap p
1	2	8.43	0.070
2	3	17.04	0.005
3	4	18.08	0.005
4	5	9.58	0.045
5	6	3.74	0.510
6	7	2.64	0.760
7	8	1.48	0.930
8	9	1.05	0.985

The break *dates* are not equally well identified. Table 3 gives residual-bootstrap confidence intervals for each break year. The Depression and mobilisation breaks are sharp, whereas the start of the long postwar regime near 1950 is estimated very imprecisely, with a percentile interval spanning 1950–1988. The lesson is that a break being statistically present does not mean its timing is pinned down to a single year.

Table 3. Bootstrap confidence intervals for the estimated break years

Break	Point year	90% interval	Std. (years)
1	1929	[1927, 1929]	0.9
2	1934	[1934, 1934]	1.2
3	1945	[1944, 1946]	3.9
4	1950	[1950, 1988]	13.6

4.3 Fiscal Context

Adding federal fiscal ratios changes the narrative from a pure growth story to a macro-fiscal story. Debt and deficit measures vary sharply across wars, recessions, and crisis periods. Same-year correlations between fiscal ratios and GDP growth are modest. In the generated panel, gross debt-to-GDP has a same-year correlation with GDP growth of about -0.32 and a one-year-lag correlation of about -0.26 . Receipts, outlays, and deficits have smaller simple correlations with annual growth.

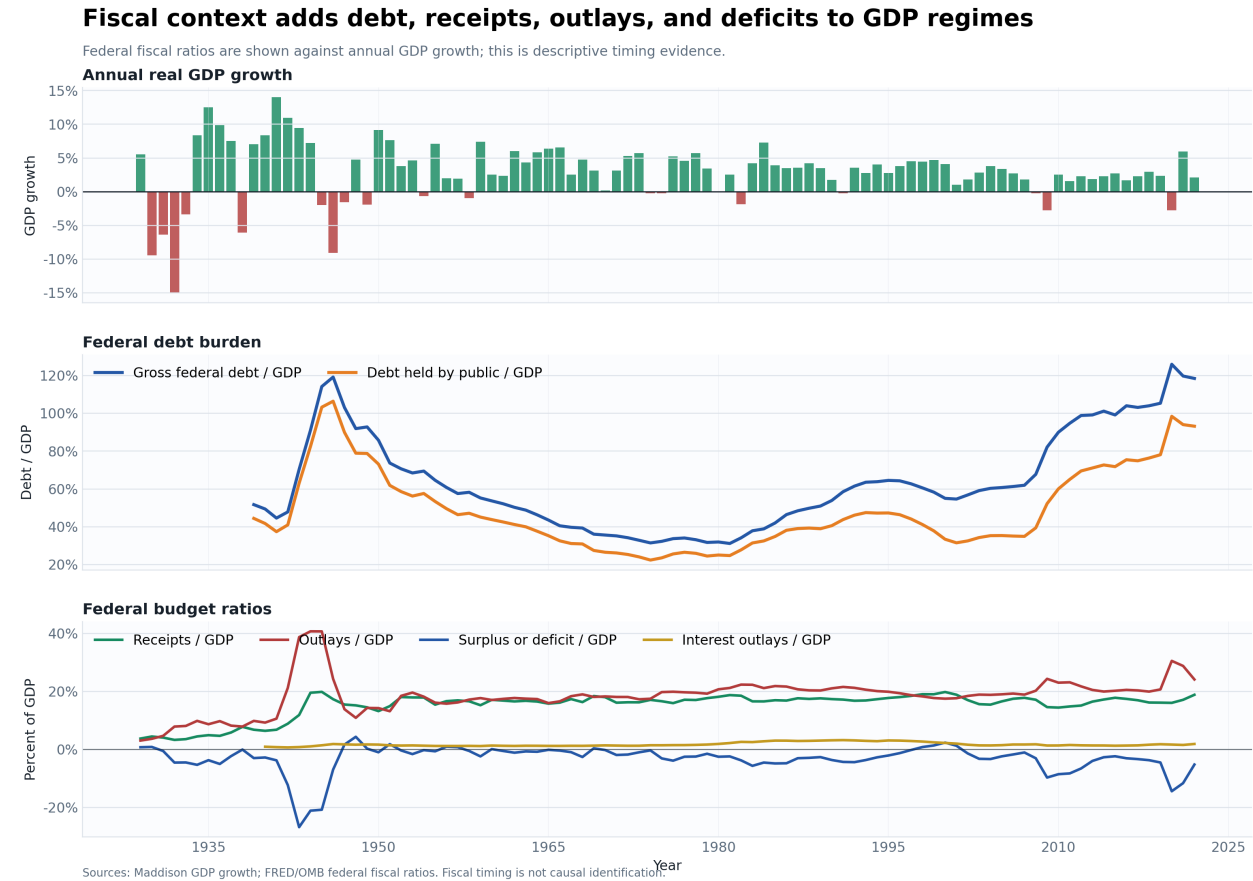


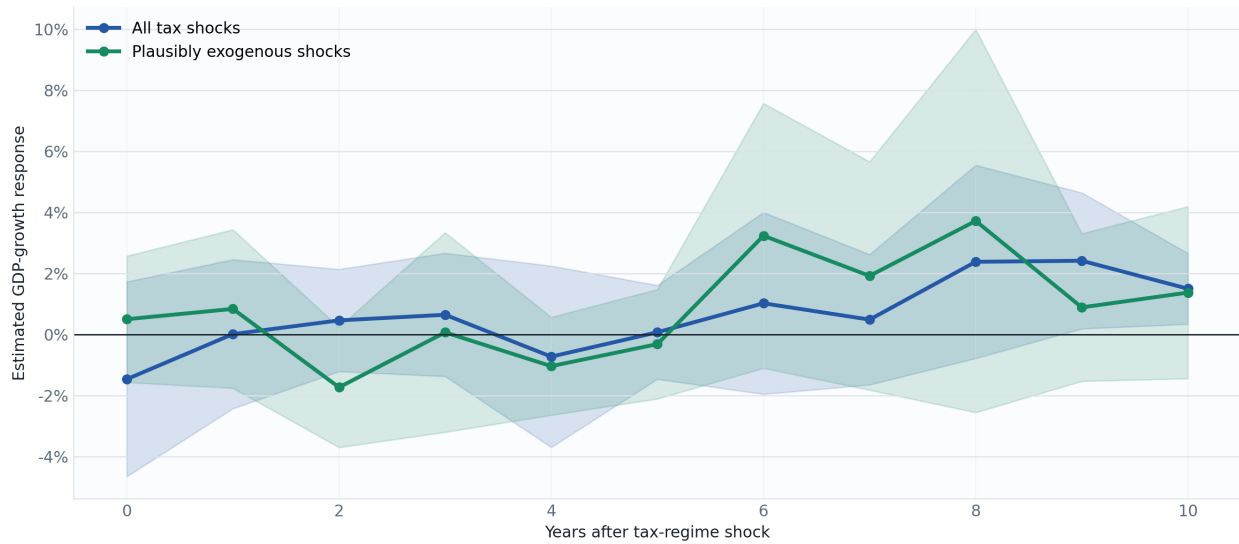
Figure 3. Federal debt, budget ratios, and annual GDP growth. The figure is a timing and context device; it does not identify fiscal causality.

4.4 Tax-Regime Dynamics

The local-projection estimates in Figure 4 show delayed paths after signed tax-regime shocks. Confidence intervals are wide, especially for the plausibly exogenous subset, because there are few major long-run reform events in the historical sample. The result should therefore be read as a dynamic screening exercise rather than a definitive tax multiplier estimate.

Tax-shock local projections show delayed GDP-growth associations

Lines estimate future annual GDP growth at each horizon after a signed tax-regime shock.



Source: Maddison GDP growth and curated tax-shock catalog. Causal interpretation is limited to plausibly exogenous shocks and remains model-dependent.

Figure 4. Local-projection estimates of GDP-growth associations after signed tax-regime shocks. The green line restricts the treatment series to events classified as plausibly exogenous long-run reforms.

The most important methodological improvement over a same-year event study is that the model allows effects to appear with delay. The data do not support a simple one-year story. Some horizons are negative, some positive, and most intervals overlap zero. That pattern is consistent with the fact that tax laws include anticipation, phase-ins, temporary provisions, and heterogeneous channels across households and firms.

5 Tax Burden Shift and Real Wages

5.1 GDP Per Capita versus Worker Buying Power

Figure 5 is central to the distributional interpretation. Using 1979 as the common index base, real GDP per capita reached about 218.4 by 2026, while real median weekly earnings reached about 113.4. Real hourly compensation (156.8) rose more than median weekly earnings, but still much less than GDP per capita. This means aggregate output per person can increase while the median worker's measured real earnings grow slowly.

GDP per capita can rise while median real earnings lag

Indexes share the same base year; the gap tracks aggregate output versus worker buying power.

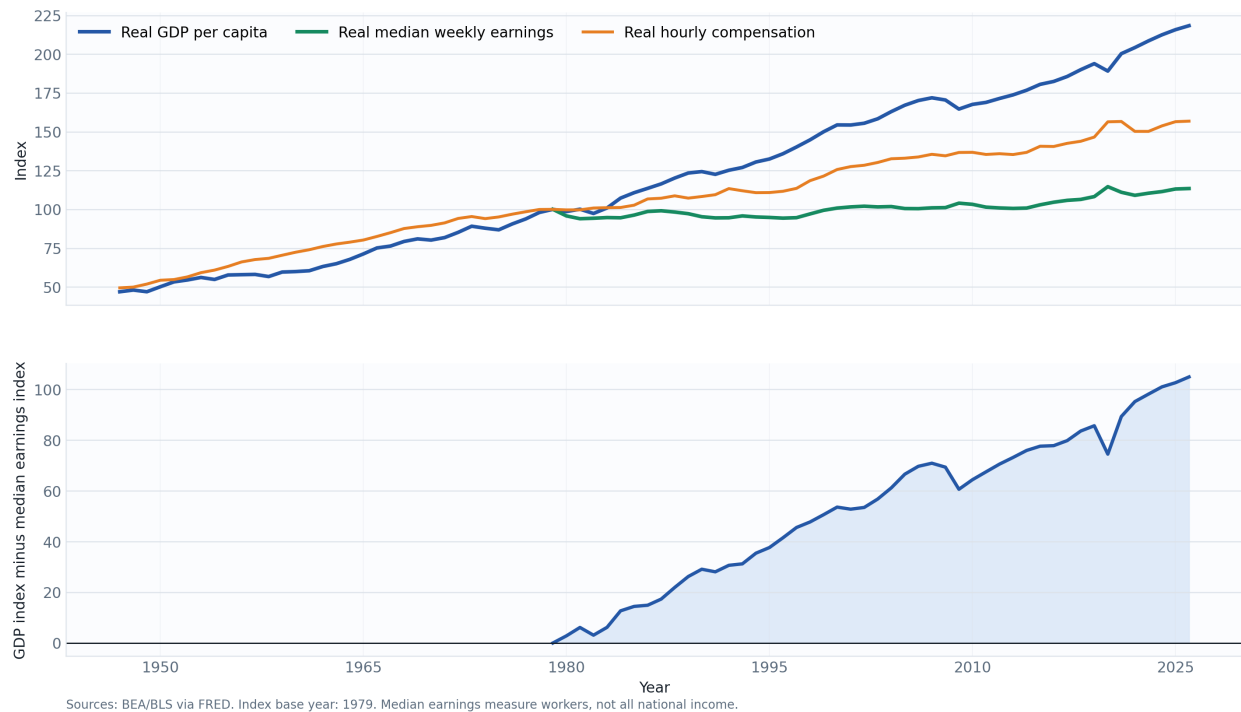


Figure 5. Real GDP per capita, real median weekly earnings, and real hourly compensation indexed to 1979. The widening gap shows why GDP growth is not equivalent to broad worker buying-power growth.

This does not mean most people necessarily became poorer in every year. It means the aggregate growth measure and the median wage-worker measure diverged substantially. A welfare analysis would also need hours worked, household composition, transfers, taxes, fringe benefits, asset income, housing costs, and consumption prices by income group.

5.2 Tax Burden Shift Proxies

Figure 6 combines three views of tax burden. The first panel shows the composition of selected federal receipts. Social-insurance contributions rose from about 15.1% of the selected federal receipt base in 1947 to about 35.1% in 2026. Corporate income taxes fell from about 29.6% to about 10.1% over the same endpoint comparison. This is consistent with a long-run movement away from corporate income taxes and toward social-insurance financing.

Tax burden shift proxies combine receipt composition and quintile tax rates

Social-insurance receipts proxy less-progressive payroll taxation; quintile rates show income-tax burden.

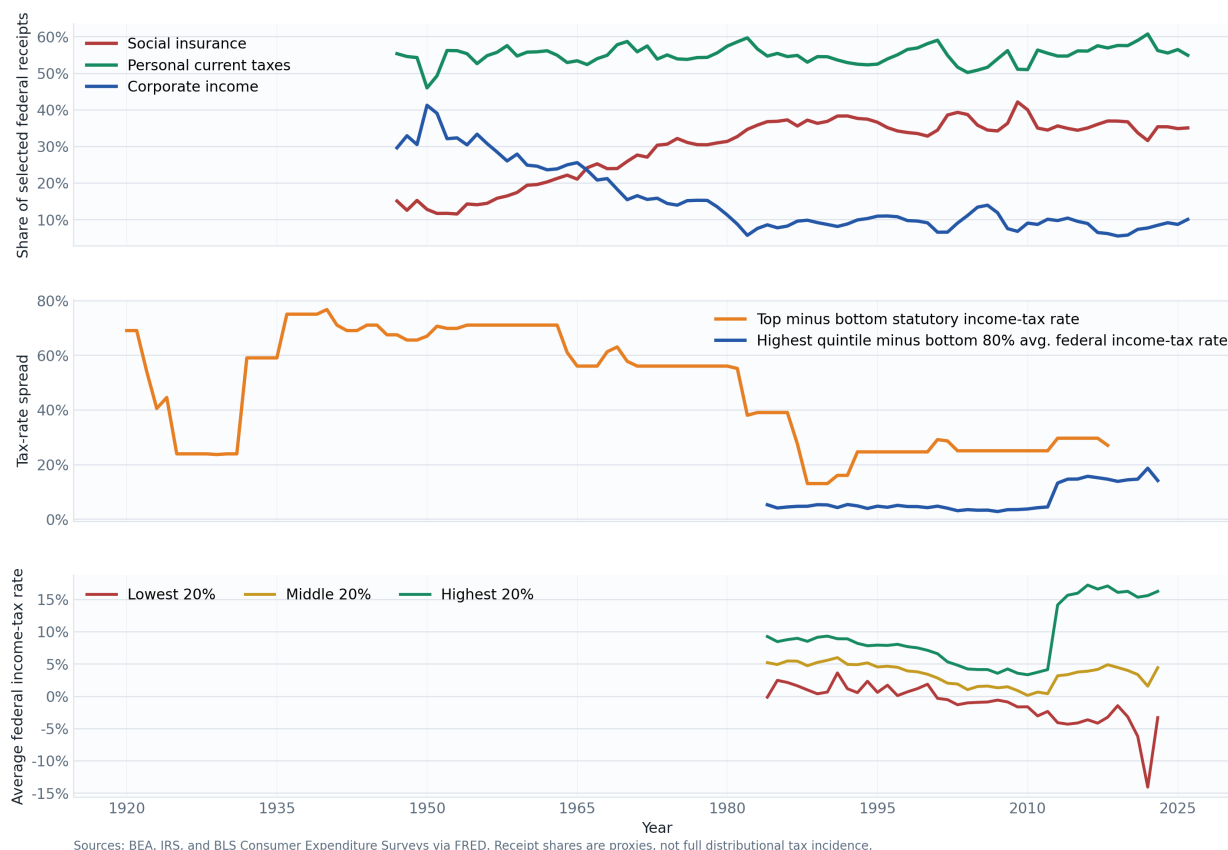


Figure 6. Tax burden shift proxies. Receipt composition captures the rise of social-insurance contributions relative to corporate income taxes; quintile rates show average federal income-tax rates by income group where Consumer Expenditure data are available.

The Consumer Expenditure quintile panel begins in 1984. In that year, the highest income quintile's federal income-tax rate exceeded the average of the bottom four quintiles by about 5.25 percentage points. By 2023, that spread was about 14.10 percentage points. This particular income-tax measure therefore does not show a simple shift of federal income-tax burden away from the top quintile. The broader burden story is more nuanced: payroll-style financing became more important, corporate income-tax receipts became less important, and income-tax progressivity by quintile remained substantial.

5.3 Association Screen

The lagged association screen regresses GDP growth, real median weekly earnings growth, and the GDP/wage gap on one-year-lagged burden-shift proxies. These are single-predictor models with HAC standard errors. The strongest pattern in the current output is not a robust GDP-growth claim; it is that the GDP/wage gap is statistically associated with some tax progressivity proxies. For example, the one-year-lag statutory top-bottom rate spread has a negative coefficient in the GDP/wage-gap regression, while the quintile rate spread has a positive coefficient. These associations should be interpreted cautiously because the models are low-dimensional and do not control for

technology, unionization, trade exposure, monetary policy, inflation, demographic composition, or sectoral change.

6 Limitations

The main limitation is identification. Breakpoints, fiscal ratios, tax laws, and wages all move with macroeconomic conditions. Recessions change tax receipts and policy decisions; wars change debt, output, and tax rates; inflation and monetary policy affect both GDP and real earnings. The tax-shock catalog partly addresses this by separating plausibly exogenous reforms from deficit, wartime, and countercyclical changes, but the classification remains historical and sample sizes are small.

The second limitation is measurement. Maddison is useful for long-run history but is not identical to BEA national accounts. Median weekly earnings are not the same as household income or total compensation. Consumer Expenditure quintile tax rates cover federal income-tax amounts, not the full tax-and-transfer system. Receipt composition is a proxy for burden shifting, not a complete incidence analysis.

The third limitation is temporal coverage. The Maddison-based GDP regime sample ends in 2022, while several modern FRED series used in the fiscal and distributional layers extend to 2026 and may include an annualized or partial latest-year observation. The report therefore reports explicit coverage by series and avoids implying that every dataset covers the same complete historical window.

7 Conclusion

The GDP regime analysis shows that the United States did not grow at one constant annual speed after 1920. A log trend summarizes the long-run level in-sample, but the level is statistically indistinguishable from a unit-root process, so stationary annual growth — read through a small number of statistically supported regimes — is the appropriate object of study. Fiscal and tax data enrich the narrative, especially by showing how debt, deficits, tax-law changes, and federal receipt composition line up with macroeconomic periods.

The distributional extension changes the substantive interpretation. Real GDP per capita and real median weekly earnings diverged strongly after 1979. That does not negate GDP growth; it clarifies who may have participated in it. From a data-science perspective, the correct conclusion is not that GDP is useless, but that GDP is incomplete. A credible assessment of economic regimes should report aggregate output, fiscal structure, tax burden proxies, and worker purchasing power together.

Reproducibility Statement

The report is generated from the repository pipeline:

- `us-gdp-regimes run --config config/default.yaml`
- `us-gdp-regimes make-fiscal-context --config config/default.yaml`
- `us-gdp-regimes make-tax-effects --config config/default.yaml`
- `us-gdp-regimes make-distributional-analysis --config config/default.yaml`

- `us-gdp-regimes export-report-numbers --config config/default.yaml`

The last command regenerates `generated_numbers.tex` and the generated tables that this report inputs, so every quoted statistic is reconciled with the model outputs. The relevant notebooks are 01 through 06 in the `notebooks/` directory. Generated raw data, model CSVs, figures, and article assets are reproducible outputs and are not committed as raw source data.

References

- [1] Bai, J. and Perron, P. Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78, 1998.
- [2] Bai, J. and Perron, P. Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1):1–22, 2003.
- [3] Jordà, Ò. Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182, 2005.
- [4] Newey, W. K. and West, K. D. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- [5] Said, S. E. and Dickey, D. A. Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71(3):599–607, 1984.
- [6] Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y. Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3):159–178, 1992.
- [7] Maddison Project Database. Maddison Project Database 2023. <https://www.rug.nl/ggdc/historicaldevelopment/maddison/>.
- [FRED GDPCA] Federal Reserve Bank of St. Louis. Real Gross Domestic Product, annual, GDPCA. <https://fred.stlouisfed.org/series/GDPCA>.
- [FRED Fiscal Ratios] Federal Reserve Bank of St. Louis. Federal fiscal ratio release tables. <https://fred.stlouisfed.org/release?rid=263>.
- [OMB Historical Tables] Office of Management and Budget. Historical Tables. <https://www.whitehouse.gov/omb/information-resources/budget/historical-tables/>.
- [IRS SOI Historical Table 23] Internal Revenue Service Statistics of Income. Historical Table 23: U.S. Individual Income Tax. <https://www.irs.gov/statistics/soi-tax-stats-historical-table-23>.
- [BLS Consumer Expenditure Surveys] U.S. Bureau of Labor Statistics. Consumer Expenditure Surveys. <https://www.bls.gov/cex/>.
- [BEA National Accounts] U.S. Bureau of Economic Analysis. National Income and Product Accounts. <https://www.bea.gov/data/gdp/gross-domestic-product>.